From Cells to Organisms

Testing the Emergence of Multicellularity

Central question. Under what physical, metabolic, ecological, developmental, and evolutionary conditions can independent cells become a persistent, integrated, self-reproducing multicellular evolutionary individual?

This expert seed defines a falsifiable simulation-and-evidence program for Viva-Genesis. It synthesizes Morowitz's emergence framework with evolutionary transitions in individuality, experimental evolution, and biophysical modeling. It distinguishes well-supported observations from hypotheses, proposes interoperable mechanistic models, and specifies a first discovery cycle capable of rejecting explanations.

Identity	Specification
Investigation ID	VG-MULTI-001
Scientific domain	Origins of biological complexity; evolutionary transitions; multicellular biophysics
Primary test systems	Snowflake yeast, predator-selected Chlamydomonas; volvocine algae and filamentous cyanobacteria as comparative systems
First deliverable	Validated minimal cell-collective simulator and claim-evidence graph
Status	Proposed seed. Tool names and manifest are design targets, not a verified importer schema.

Executive scientific rationale

The existence of adherent cells is not sufficient to establish a major transition. A defensible investigation must identify a sequence of separately measurable properties: collective formation, physiological coordination, reproducible group reproduction, heritable group-level traits, and adaptation at the collective level. These properties may evolve in different orders and need not all appear in every lineage.

Discriminating objective: determine when size-dependent ecological selection and elementary adhesion explain observed group formation, and when metabolism, development, heritable life cycles, or conflict suppression are necessary for persistence and further complexity.

1. Conceptual foundations and physical constraints

Constraint	Mechanistic implication	Testable measurement
Adhesion and topology	Group stability depends on cell-cell bond formation, geometry and fragmentation.	Bond lifetime, group connectivity, propagule distributions
Transport and metabolic demand	Increasing group size can impose nutrient, oxygen and waste gradients.	Growth/viability versus radius, diffusion length, uptake flux
Ecological selection	Predation and settling can reward size even without specialization.	Fitness advantage under predator/settling treatments
Coordination and differentiation	Signals, local gradients and regulatory plasticity may create distinct cell states.	Spatial fate maps, group productivity, phenotypic stability
Heritability and conflict	Within-collective competition can undermine group-level adaptations.	Between/within-group selection, life-cycle fidelity

Physical scalings used as baseline checks: for a sphere, surface-to-volume ratio $A/V = 3/R$; a characteristic diffusion time scales as L^2/D. These relations constrain, but do not uniquely predict, colony dimensions because shape, metabolism, convection and active transport alter outcomes.

2. Scientific questions

ID	Question and desired discrimination
Q1	Which adhesion, reproductive and ecological regimes create persistent collectives rather than transient aggregates?
Q2	How do oxygen, nutrients, cell shape, packing and geometry limit collective size?
Q3	When can metabolite exchange and functional specialization increase group fitness?
Q4	Can environmental gradients create stable cell-state differentiation without a dedicated developmental program?
Q5	When do reproduction, heritability and conflict suppression shift evolutionary adaptation to the collective level?
Q6	Which combinations of constraints explain why complex multicellularity is less common than simple grouping?

3. Hypotheses, claims, and falsification

ID	Competing hypothesis	Discriminating prediction
H1	Ecology-first: size-dependent survival or settling favors initial groups.	Under selected ecological regimes, simple adhesive collectives outperform unicells without metabolic exchange.
H2	Metabolism-first: cooperation and resource partitioning promote integration.	Metabolically complementary groups gain fitness even when size protection is controlled.
H3	Plasticity-first: local gradients induce divergent cell states.	Spatial fate patterns arise before evolved cell-fate circuitry; regulatory fixation increases stability.
H4	Alignment-first: clonality and group reproduction enable stable collectives.	Clonal bottlenecked populations maintain group adaptations more effectively than genetically mixed groups.
H5	Coupled constraints: no single mechanism is generally sufficient.	Different combinations create distinct viable regimes, with interactions exceeding additive effects.

Evidence-linked claims (seed status)

Claim	Initial assessment	Required challenge
C1 Predation can favor groups	Direct experimental evidence [R4]	Compare predator-free controls and alternative ecological effects.
C2 Clonality limits conflict	Strong theory/comparative rationale [R2]	Test mixed founders and somatic cheaters.
C3 Diffusion limits size	Physical premise and evidence [R3,R5]	Control shape, transport and metabolic regime.
C4 Mechanics permits macroscopic clusters	Direct yeast evidence [R5]	Compare explicit geometry/mechanics models.
C5 Gradients can initiate specialization	Mechanistic hypothesis	Demonstrate specialization exceeding a non-regulated null.
C6 Life cycles sustain group selection	Evolutionary theory and yeast data [R2,R6]	Measure propagule inheritance and multilevel selection.
C7 Complexity requires integration	Broad comparative inference [R3]	Avoid asserting a universal necessary pathway.

4. Model architecture and simulation interfaces

Use a backend-neutral biological blueprint compiled into process-bigraph composites. Every simulator receives the same canonical cell and collective states, while mechanics and intracellular physiology can be swapped without changing the question being investigated.

Component	Minimal input/output contract	Candidate implementation
Cell growth/division	Nutrients, state → biomass, births, lineage edges	Custom agent process; PhysiCell
Mechanics/adhesion	Geometry, bonds, division → positions, contact graph, breakage	Spatio-Flux/Pymunk; viva-cpm; vivarium-tyssue
Transport fields	Cell uptake/secretion → spatial concentration fields	Spatio-Flux PDE; Morpheus; CompuCell3D
Metabolism	Nutrient boundary conditions → growth/yields/exchange	COBRApy/dFBA; COMETS
Cell-state regulation	Signals/metabolites → phenotype state transitions	COPASI, Tellurium, Boolean/ODE modules
Evolution/life cycle	Heritable traits, propagules, outcomes → next generation	Custom population-genetic process
Composition	Typed shared stores, scheduler, provenance, recording	Vivarium / Process Bigraph

Canonical entities. Cell: id, parent, lineage, position, geometry, biomass, resource state, phenotype, genotype, bonds. Collective: id, member graph, total biomass, geometry, nutrient field exposure, viability, propagule production, ancestry, trait distribution. Environmental state: resources, oxygen, predators, flow/settling field. All outputs should record units, parameter provenance, random seed and backend version.

Cross-backend verification. Begin with identical simple tests for growth, contact, conservation and fragmentation. Compare summary observables rather than raw trajectories; use matched initial conditions and stochastic replicate ensembles. Parameterize model-form uncertainty explicitly.

5. First six studies

Study	Purpose and minimal design	Decision criterion
S01 Formation	Contrast clonal incomplete separation, aggregation and reversible adhesion. Sweep bond strength, division and fragmentation.	Persistent group life cycle and heritable cluster traits versus non-group null.
S02 Ecological selection	Add size-dependent predation, settling and no-selection controls.	Ecological benefit explains group emergence without added physiology.
S03 Transport limits	Couple oxygen/nutrient/waste PDEs to growth across colony geometries and metabolic regimes.	Predict size/viability changes across experimental regimes.
S04 Morphomechanics	Vary cell aspect ratio, branch topology, packing and bond strength; compare mechanics backends.	Predict toughness and propagule-size distributions.
S05 Differentiation	Enable metabolite exchange and inducible cell fate, with no-fate and random-fate nulls.	Specialization improves out-of-sample group performance.
S06 Evolutionary individuality	Compare single-cell bottlenecks, multicellular propagules and mixed-genotype groups with cheating.	Heritable between-group fitness gains remain stable across generations.

6. Study-level evidence, benchmarks and decision rules

S01 and S02 benchmark against Ratcliff et al. (2012) and Herron et al. (2019): group emergence, settling and predator effects. S03 and S04 benchmark against Bozdag et al. (2023): oxygen conditions, cell elongation, macroscopic size and mechanical robustness. S05 uses volvocine and cyanobacterial differentiation as qualitative comparisons unless directly usable datasets are ingested. S06 evaluates group heritability and within-/between-group selection; historical reconstruction is not implied by present-day experiments.

Explicit stopping and promotion rules. A model can be promoted only after reproducing baseline experimental trends, satisfying conservation/unit checks and predicting at least one held-out observation or cross-condition perturbation. Failed models remain linked to their falsifying evidence. Where multiple models fit equally well, the engine should report non-identifiability and select a discriminating intervention, not announce discovery.

7. Discovery-engine graph objects and cycle

Entity	Required graph relationships
Research question	asks → claims; motivates → studies
Hypothesis / claim	predicts → observation; supported_by / challenged_by → evidence
Resource	contains → dataset/model/figure; provenance and DOI/URL
Model / process	implements → mechanism; produces → simulation run
Study / run	tests → hypothesis; uses → model/dataset; yields → observation
Validator	checks → numerical integrity / empirical fit / causal identification / provenance
Narrative	summarizes → supported claims, unresolved conflicts, next experiment

Discovery cycle: Ingest resources → extract claims and quantitative evidence → map contradictions and gaps → instantiate competing models → validate and calibrate → generate predictions → compare to held-out evidence → revise claim credibility → rank next experiments by expected discrimination and computational cost → regenerate the narrative.

First-cycle priority

Test whether ecological selection plus simple clonal adhesion suffices to reproduce group formation, fragmentation and group-level fitness advantages. Compare at least three mechanistic models: (A) adhesion only, (B) adhesion + predation/settling ecology, (C) model B + oxygen-limited growth. Hold out at least one ecological or metabolic regime for prediction. Use sensitivity analysis and replicate stochastic seeds.

8. Resource ingestion manifest

The resources below are intended to be added as distinct library objects, with DOI, abstract, evidence-type tags, and claim links. Bibliographic details and accessible supplementary data should be checked at ingestion. Morowitz chapter-level interpretation is a conceptual synthesis, not a verbatim extraction from the copyrighted text.

R1 · Morowitz, H. J. (2002). The Emergence of Everything. Chapter 14: Multicellularity.

Foundational emergence narrative; theoretical book chapter.
https://academic.oup.com/book/52761

R2 · Grosberg, R. K. & Strathmann, R. R. (2007). The evolution of multicellularity: a minor major transition?

Clonality, conflict, individuality; conceptual review.
https://doi.org/10.1146/annurev.ecolsys.36.102403.114735

R3 · Knoll, A. H. (2011). The multiple origins of complex multicellularity.

Comparative origins, transport limits, complexity; review.
https://doi.org/10.1146/annurev.earth.031208.100209

R4 · Herron, M. D. et al. (2019). De novo origins of multicellularity in response to predation.

Predation-selected Chlamydomonas experimental evolution.
https://doi.org/10.1038/s41598-019-39558-8

R5 · Bozdag, G. O. et al. (2023). De novo evolution of macroscopic multicellularity.

Experimental evolution; mechanical traits; public source data/code.
https://doi.org/10.1038/s41586-023-06052-1

R6 · Ratcliff, W. C. et al. (2012). Experimental evolution of multicellularity.

Yeast settling selection, life cycle, group reproduction.
https://doi.org/10.1073/pnas.1115323109

R7 · Kirk, D. L. (2005). A twelve-step program for evolving multicellularity and a division of labor.

Volvocine progression and development; comparative account.
https://doi.org/10.1002/bies.20197

R8 · Bozdag et al. (2023), source data and code repository.

Figure data, analysis code, 3D model and sequencing links.
https://github.com/ozanbozdag/de_novo_evolution_of_macroscopic_multicellularity

R9 · NIH SRA, PRJNA943273.

Sequencing from macroscopic multicellularity experiment.
https://www.ncbi.nlm.nih.gov/sra/?term=PRJNA943273

9. Minimal import specification (proposed)

Investigation id: VG-MULTI-001; title: From Cells to Organisms: Testing the Emergence of Multicellularity; status: seeded. Create objects Q1–Q6, H1–H5, C1–C7, S01–S06 and R1–R9, preserve cross-links, and mark every provisional claim as unvalidated until mapped to source-specific evidence. Register model families: agent-based, particle mechanics, Cellular Potts, vertex, reaction-diffusion, dFBA, regulatory, and evolutionary population models.

Seed acceptance checklist: (1) each claim has an evidence status; (2) every study has at least one falsifiable prediction and a baseline/null; (3) resource URLs resolve; (4) physical units and model interfaces are explicit; (5) no conclusion is declared established merely from simulation fit; (6) competing hypotheses remain visible; (7) outputs, failures and provenance can be stored in the investigation graph.

10. Expected discovery products

- An evidence-linked narrative of origins of multicellularity, updated through simulation cycles.
- A reusable suite of interoperable cell/collective model blueprints and backend comparisons.
- A phase atlas of viable, unstable and selectively advantageous multicellular regimes.
- Quantitative boundaries separating physical aggregation, physiological integration and evolutionary individuality.
- A prioritized queue of decisive simulations and experimentally testable predictions.

Interpretive caution. These studies test mechanisms and constraints. Reproducing multicellular phenotypes does not by itself establish the actual prehistoric sequence of evolutionary events. The engine should explicitly preserve distinctions among possibility, sufficiency, necessity, and historical inference.